

Lab 08 Prompts: Test and Debug the API

Course: Create RESTful APIs and Web Apps with Python Flask Course code: TGS-2021010365 Version: v11.0
Maps to: A5, A6, A7

Prompt execution contract

Use `python ai_vibe.py --dry-run` to inspect the exact request. The script uses the OpenAI Python SDK Responses API with a Pydantic CodeBundle. It writes drafts only to `working-files/generated/`; it never overwrites trusted source. Keep `OPENAI_API_KEY` in the shell, not in prompts, files, screenshots or browser JavaScript.

Learner prompt - copy exactly for the first run

You are working on Lab 08 of a Northstar Commerce API. Using the seeded failing cases in `mock-data.json`, produce a focused pytest regression suite and a debug plan. Reproduce the failure with the smallest request, assert response and persisted state, identify the likely boundary, and propose the minimum patch without weakening existing acceptance tests.

Use only the synthetic context in `mock-data.json` and the current project files supplied by `ai_vibe.py`. Preserve existing API behaviour unless this prompt explicitly changes it. Use Python 3.11+, FastAPI, Pydantic v2, SQLite parameter binding and pytest/TestClient. Do not include secrets, database files, caches or invented test results. Return a typed CodeBundle containing complete file contents for only these targets:

- `tests/test_regression.py`
- `debug-plan.md`

For every generated file, state its purpose. Include assumptions and concrete verification commands. The learner will review the diff and run the commands before accepting any code.

Evidence-led refinement prompt

The previous generated bundle did not yet pass the acceptance check below. Use the pasted failure evidence, identify the smallest contract or implementation mismatch, and return a revised CodeBundle containing only files that must change. Do not weaken or delete a passing test.

Acceptance check: The seeded test fails before the fix; the full suite passes after it; the regression assertion remains.

Paste exact evidence here:

HTTP request/response, traceback, assertion diff or log request ID

Review checklist

- Confirm every generated path is inside `working-files/generated/`.
- Reject hard-coded keys, personal data, f-string SQL and unrestricted CORS.
- Compare generated behaviour with the written API contract and mock data.
- Run `pytest -q`; inspect status, JSON and SQLite state, not only the exit code.
- Record prompt, model name, assumptions, accepted diff and final evidence.